

3W AUDIO AMPLIFIER Using TDA7056

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Audio amplifier circuits commonly use single-chip audio ICs like LM384 and LM386. These circuits look very easy to assemble but require additional components like resistors, capacitors and sometime inductors. Here we present the circuit of a 3W single-chip audio amplifier using TDA7056 that only requires two capacitors for biasing and one potmeter for volume control (and that's also op-

tional). You also need to connect a 3W, 16-ohm (or 1W, 8-ohm) speaker at the output of the circuit. So when an audio signal is applied to the input, you can hear the amplified output from the speaker. The author's prototype is shown in Fig. 1.

The main features of TDA7056 include:

1. Bridge-tied load working principle
2. Supply voltage range of 3V to 18V
3. Gain of 40dB

4. No external components required
5. No switch-on or switch-off clicks
6. Good overall stability
7. Low power consumption

Working on bridge-tied load principle, the circuit can output 1W power into an 8-ohm load with a power supply of 6V, and 3W power into a 16-ohm load with a power supply of 12V.

Circuit and working

Fig. 2 shows circuit of the 3W audio amplifier using TDA7056 (IC4). It has two sections: power supply and amplifier.

Power supply. Here the AC mains power supply is stepped down using transformer X1, rectified by a full-wave rectifier comprising diodes D1 and D2, filtered by capacitor C1 and fed to 5V voltage regulator 7805 (IC2) and 12V voltage regulator 7812 (IC3) to maintain constant +5V DC supply for melody generator UM66 (IC3) and +12V DC supply for audio amplifier TDA7056 (IC4), respectively.

Amplifier. The amplifier section is built using two chips: melody

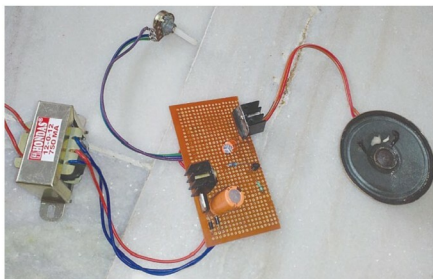


Fig. 1: Author's prototype

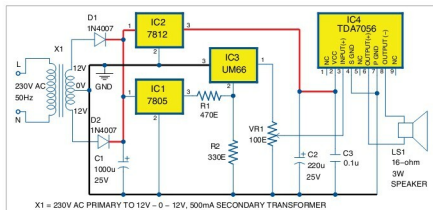


Fig. 2: Circuit of 3W audio amplifier using TDA7056

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V regulator
- IC2 - 7812, 12V regulator
- IC3 - UM66 melody generator
- IC4 - TDA7056 audio amplifier
- D1, D2 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 470-ohm
- R2 - 330-ohm
- VR1 - 100-ohm potmeter

Capacitors:

- C1 - 1000 μ F, 25V electrolytic
- C2 - 220 μ F, 25V electrolytic
- C3 - 0.1 μ F ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 12V-0-12V, 500mA secondary transformer
- LS1 - 16-ohm, 3W loudspeaker
- Heat-sink for IC1, IC2 and IC4
- Mains power cord